ie UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

BACHELOR IN DATA AND BUSINESS ANALYTICS

TIME SERIES ANALYSIS

Session 7 · Holt and Holt-Winters

Professor: Angel Mora Segura

07 October 2026 · 18:00–19:30 · Live in-person



COURSE SCHEDULE · 20 SESSIONS · 80 MINUTES EACH

Session by Session

1	02 Sep	Introduction · R setup · forecasting as a business problem ✓
2	03 Sep	Time plots, seasonal plots, EDA ✓
3	23 Sep	Lags, ACF, PACF ✓
4	24 Sep	Decomposition, STL, Box-Cox · Checkpoint 1 ✓
5	30 Sep	Benchmarks: Naive & Seasonal Naive ✓
6	01 Oct	Simple exponential smoothing ✓
7	**07 Oct**	**Holt, damped trend, Holt-Winters** **TODAY**
8	08 Oct	ETS taxonomy + model selection
9	15 Oct	**Midterm (intermediate test) — 20% · Checkpoint 2**
10	19 Oct	AR / MA
11	22 Oct	ARMA + stationarity
12	28 Oct	ARMA + integration + diagnostics
13	29 Oct	ARIMA
14	12 Nov	Seasonal ARIMA (SARIMA)
15	18 Nov	Automatic ARIMA + model selection
16	19 Nov	Complex seasonality / Fourier
17	25 Nov	Integrated Forecasting Workshop · Checkpoint 3
18	02 Dec	**Group presentations — 30% · double session, part 1**
19	02 Dec	**Group presentations — 30% · double session, part 2**
20	11 Dec	**Final exam in R — 40%**

TODAY'S SESSION

Following Trend and Season

Holt's linear trend, the damped trend that practitioners prefer, and the Holt-Winters seasonal method in its additive and multiplicative forms.

FPP3 · Sections 8.2 – 8.3

TYPICAL CLASS STRUCTURE · 80 MINUTES

The Classroom Is the Laboratory

The same three blocks every session. The work on your case happens here, with me in the room.

~30 MIN

BLOCK 1

Concepts & R application

The methodology, its intuition and its assumptions — aimed at **what problem this technique solves** — then straight into its practical implementation in R.

You leave with code that runs.

~30 MIN

BLOCK 2

Business Case Lab

Teams apply or evaluate the methodology in the context of their own case. Finished code is not the requirement.

Output can be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or a justified argument for why this technique does not apply.

~20 MIN

BLOCK 3

Business Case Clinic

Selected teams share their approach, findings or difficulties in about 5 minutes each.

The class discusses alternatives and gives feedback; I close the loop.

Block durations vary with the topic: technically demanding sessions take more time for explanation and R; later sessions shift progressively towards project development and feedback.

BY THE END OF TODAY YOU SHOULD BE ABLE TO

Session Objectives

1. Extend the level equation with a **trend** component, and explain what β controls.

2. Explain why a **damped trend** is usually safer than a linear one at long horizons, and what ϕ does.

3. Add a **seasonal** component and choose between the additive and multiplicative Holt-Winters forms.

4. Fit all of them in R and compare them on the **same test set** as your benchmark.

5. **Decide for your case** whether trend, damping and seasonality each earn their place.

BLOCK 1 · 30 MINUTES

Concept + R Application

Three components, three smoothing parameters, one family of methods.

IN CLASS: THE CORE IDEAS · ANNEX + QUARTO: EQUATIONS, R DETAILS AND THE REST

Building Up From the Level

Each method today adds one component to SES, and each component gets its own smoothing parameter. The logic is identical every time: blend the newest evidence with the previous estimate.

SES · SESSION 6

Level *l*

Where the series is now. Smoothed with α.

HOLT

Level + trend *b*

Where it is, and how fast it is moving. The trend is smoothed with β.

DAMPED HOLT

Level + damped trend

The trend is allowed to fade out over the horizon at a rate ϕ.

HOLT-WINTERS

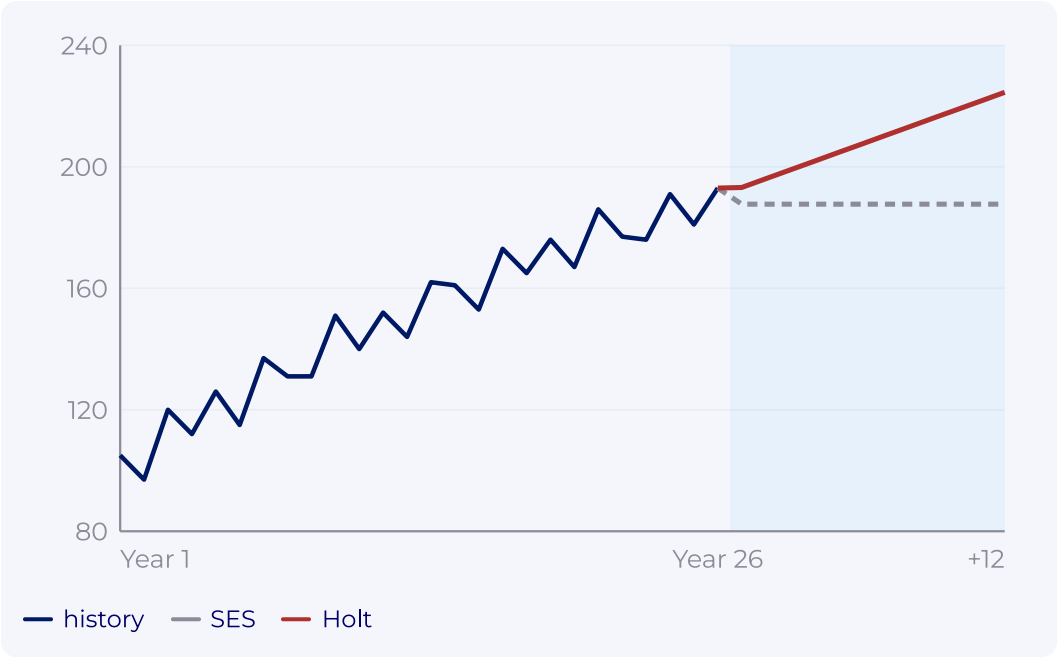
Level + trend + season *s*

Adds a seasonal pattern of period m, smoothed with γ.

Use Block 1 to choose: a clear trend in the STL panel → add *b*. A seasonal spike family in the ACF → add *s*. Neither → stay with SES.

EXAMPLE A GROWING SERIES WHOSE GROWTH IS SLOWING

What a Trend Component Changes

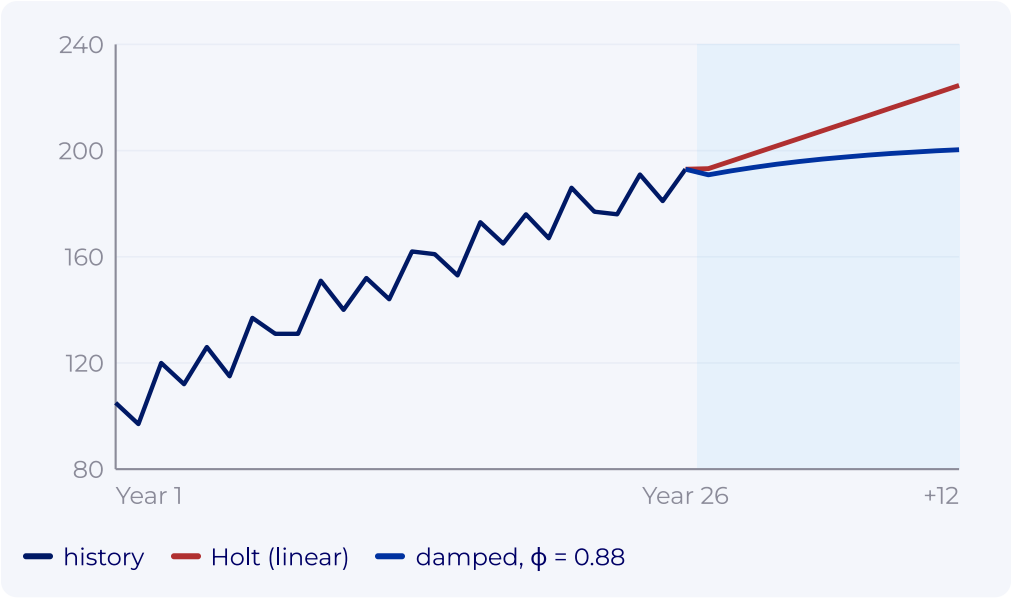


SES stops; Holt keeps climbing.

- SES forecasts the current level for every horizon. On a growing series, it is wrong from the first step and more wrong at every step after.
- Holt extends the recent slope. Twelve periods out it forecasts about 225.
- But look at the history: growth has been **slowing** for years. A straight line ignores that — which is exactly the problem damping solves.

The Damped Trend

A parameter ϕ between 0 and 1 shrinks the trend a little more at every step ahead. Short-range forecasts still follow the slope; long-range ones level off. Empirically, it is one of the most reliable ideas in forecasting.



$$\hat{y}_{t+h|t} = l_t + (\phi + \phi^2 + \ldots + \phi^h)\, b_t$$

- **ϕ = 1** is Holt. **ϕ = 0.8–0.98** is the usual range. Below 0.8 the damping is so strong the trend barely matters.
- Twelve steps out: 225 (linear) against 200 (damped).
- Why it works: trends in business data rarely continue at the same rate. Damping is a built-in bet that they will slow down.

Additive or Multiplicative Seasonality?

The same question you answered by eye in Session 2 and formally in Session 4. Holt-Winters makes you answer it a third time — and now it changes the forecast.

Additive

$$\hat{y} = (l + hb) + s$$

Seasonal effects in the units of the data: “July adds 2,000 units”. Right when the seasonal swing stays the same width as the level changes.

Multiplicative

$$\hat{y} = (l + hb) \times s$$

Seasonal effects as proportions: “July is 40% above trend”. Right when the swing grows with the level — the usual case in a growing business.

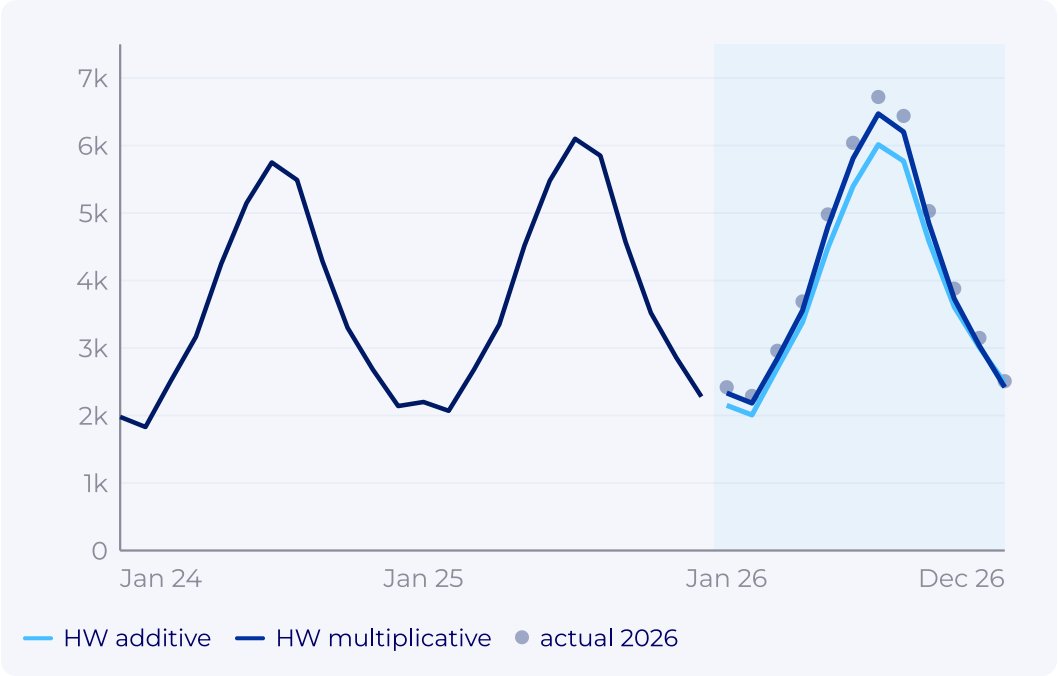
Two equivalent routes

- Multiplicative Holt-Winters on the original data.
- Additive Holt-Winters on the **log** of the data — using the transformation you chose in Session 4.
- Both usually give similar forecasts. Try both, and let the test set decide.

Same diagnosis as Session 2: if the seasonal band widens with the level, use the multiplicative form — or take logs first.

EXAMPLE HAPPY COW · SAME SPLIT AS SESSION 5

Holt-Winters on the Running Series



The multiplicative version follows the 2026 peak.

- The series grows, and its summer peak grows with it. That is the multiplicative pattern you diagnosed in Session 4.
- Additive Holt-Winters adds a fixed seasonal amount, so it under-shoots the peak and over-shoots the trough.
- Multiplicative scales the season to the level and lands close to every 2026 month.

EXAMPLE TEST SET · 2026

The Table Now Has Four Rows

Method	RMSE	MAE	MAPE	MASE
HW multiplicative	**167**	**157**	**3.8%**	**0.64**
HW damped mult.	363	348	8.7%	1.42
HW additive	430	376	8.6%	1.54
Seasonal naive	411	386	9.3%	1.57

Multiplicative Holt-Winters is the first model to beat the benchmark clearly.

- MAE falls from 386 (seasonal naive) to 157 — about 59% better, and MASE drops below 1.
- Additive Holt-Winters barely improves on the benchmark: the wrong seasonal form costs almost all of the gain.
- The damped version does worse here, because this series's growth did **not** slow in 2026. Damping is a bet, and on this test period it lost.

R APPLICATION

Every Method, One Pipeline

```
fit <- train |>
  model(
    snaive  = SNAIVE(units),
    holt    = ETS(units ~ error("A") +
                  trend("A") + season("N")),
    damped  = ETS(units ~ error("A") +
                  trend("Ad") + season("N")),
    hw_add  = ETS(units ~ error("A") +
                  trend("A") + season("A")),
    hw_mult = ETS(units ~ error("M") +
                  trend("A") + season("M"))
  )
fit |> forecast(h = 12) |>
  accuracy(sales) |> arrange(MASE)
```

Reading the specification

- Each method is an ETS model with a different setting for each component: **N** none, **A** additive, **Ad** additive damped, **M** multiplicative.
- Holt is `trend("A")`; damped Holt is `trend("Ad")`; Holt-Winters adds `season("A")` or `season("M")`.
- `report()` on any of them prints α, β, γ and, for damped models, ϕ.
- Tomorrow, this notation becomes the whole ETS family — and R learns to choose the settings for you.

Every method from Sessions 6 and 7 is one line inside `model()`: change the three letters, keep the pipeline.

Which Method for Which Series

Read the answers from your Block 1 diagnostics. Then fit the candidates this table points to — not every method — and let the test set decide between them.

WHAT BLOCK 1 SHOWED	CANDIDATES	WHY
No trend, no season	**SES**	Nothing to extrapolate; smooth the level.
Trend, no season	**Holt, damped Holt**	Damped by default; linear only if the test set insists.
Season, stable width	**Holt-Winters additive**	Seasonal swing does not grow with the level.
Season, width grows	**Holt-Winters multiplicative, or additive on log**	The proportional season you diagnosed in Session 4.
Trend and season, horizon long	**Damped Holt-Winters**	Protects long-range forecasts from an over-extended trend.

BLOCK 2 · 30 MINUTES

Business Case Lab

Fit the candidates your diagnostics point to, add them to the table, and see whether any of them beats the benchmark.

WHAT TO EXPECT IN EACH CASE

Holt-Winters and Your Case

CASE 1 · RETAIL

Happy Cow Ice Cream

Holt-Winters with m = 7 captures the weekend effect. Try additive and multiplicative; with temperature-driven swings, multiplicative often wins.

Watch for: the annual season is not in seven months of data. Holt-Winters will only model the weekly one.

CASE 2 · AIRLINES

Balearic Airlines

Weekly seasonality, little trend across one year: Holt-Winters without trend (`trend("N")`) is a sensible candidate.

Watch for: holidays land in the seasonal indices and distort them for every week. Compare against STL + SES from Session 6.

CASE 3 · MANUFACTURING

Clean Air Limited

Monthly, m = 12, with 24 training months — the bare minimum. Damped Holt-Winters is the natural candidate for a year-ahead commitment.

Watch for: with two cycles, γ is almost unidentifiable. If the fit looks unstable, fall back to STL + damped Holt.

CASE 4 · E-COMMERCE

Wayfair

Holt-Winters works at the Decor category level. At SKU level, the long runs of zeros rule out the multiplicative form.

Watch for: multiplicative seasonality is undefined when demand is zero. Use additive, or model aggregated levels only.

BLOCK 3 · 20 MINUTES

Case Clinic

Three or four teams: which components did your series need, and did they pay off?

SESSION 7 RECAP

What to Take Away

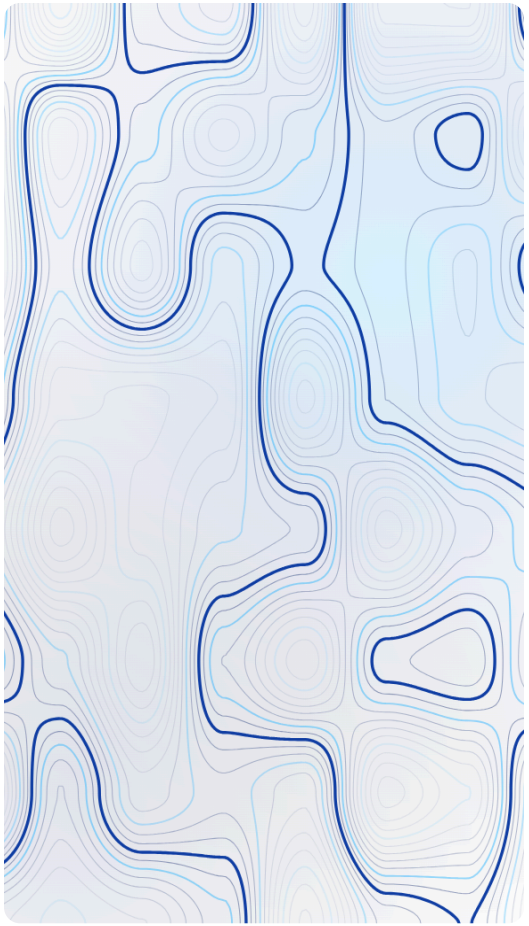
One component at a time. Level (α), trend (β), season (γ) — each updated with the same blending logic.

Damp the trend by default. A linear trend extrapolates today's growth forever; damping bets that it slows.

Match the seasonal form to the data. Additive if the swing is constant, multiplicative if it grows with the level.

Fit what the diagnostics ask for. Choose candidates from Block 1, then let the test set decide.

Tomorrow: all of these become one family, ETS. R chooses among them with an information criterion — and you learn when to trust that choice.



END OF SESSION 07

TIME SERIES ANALYSIS

Holt and Holt-Winters

NEXT · SESSION 8 · 08 OCTOBER

ETS taxonomy and model selection

Every smoothing method you have seen becomes one member of a single family — and R chooses between them for you.

The annex after this slide holds the reference material, including the R cheat sheet on slide 29.

Angel Mora Segura · IE University

07 October 2026

ANNEX · REFERENCE MATERIAL

For Study, Not for Class

Equations, R details, the cheat sheet and the common mistakes. Everything here is also developed in the session’s Quarto document.

Holt's Linear Trend Method

The level equation from Session 6, plus a second equation that tracks the slope. The forecast is no longer flat: it extends the current level along the current trend.

$$\text{Forecast: } \hat{y}_{t+h|t} = l_t + h\,b_t$$

$$\text{Level: } l_t = \alpha y_t + (1-\alpha)(l_{t-1} + b_{t-1})$$

$$\text{Trend: } b_t = \beta(l_t - l_{t-1}) + (1-\beta)b_{t-1}$$

- **The level** now blends the new observation with where the previous level *plus* the previous trend said we would be.
- **The trend** blends the latest change in level with the previous trend estimate. β close to 0: a stable, slowly updated slope. β close to 1: the slope follows every change.
- **The forecast** is a straight line: level plus h times the trend. It keeps going forever, which is its main weakness.

Holt = SES + one equation for the slope. The damped version, next, changes only the forecast line.

Why Forecasters Prefer the Damped Trend

A linear trend is a strong claim: that today's growth rate will hold for the whole horizon. The damped trend makes a weaker, more defensible one.

EVIDENCE

It wins in large competitions

Across thousands of real series in forecasting competitions, damped trend methods have been among the most accurate automatic methods, especially at longer horizons.

RISK

It fails more gracefully

If the trend does continue, damping under-forecasts moderately. If the trend stops and you did not damp, you over-forecast by a lot — and that error grows every period.

BUSINESS

It matches how growth behaves

Markets saturate, products mature, capacity binds. A trend that never slows is rarely what a manager actually believes.

Default rule: if you add a trend, try the damped version too, and keep the linear one only if the test set clearly prefers it.

Holt-Winters: Adding the Season

A third equation tracks a seasonal index for each of the m seasons. Each index is updated only when its season comes round again, so a monthly model updates the January index once a year.

Additive form

$$\hat{y}_{t+h|t} = l_t + h\,b_t + s_{t+h-m}$$

$$l_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(l_{t-1} + b_{t-1})$$

$$s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m}$$

The trend equation is unchanged from Holt.

- **The level** is updated on the *deseasonalised* observation: y minus its seasonal index.
- **The season** blends what this period says about its seasonal effect with the index from the same season last cycle.
- **γ close to 0**: the seasonal shape is almost fixed. **γ larger**: the shape is allowed to evolve — like the STL season window in Session 4.
- Three parameters, α, β and γ, all estimated from the data.

γ plays the role of the STL season window from Session 4: small γ, a stable shape; large γ, a shape that evolves.

EVERYTHING FROM TODAY, IN ONE PLACE

Session 7 R Cheat Sheet

```
# trend methods
holt   = ETS(y ~ error("A") +
             trend("A") + season("N"))
damped = ETS(y ~ error("A") +
             trend("Ad") + season("N"))
# seasonal methods
hw_add  = ETS(y ~ error("A") +
              trend("A") + season("A"))
hw_mult = ETS(y ~ error("M") +
              trend("A") + season("M"))
hw_damp = ETS(y ~ error("M") +
              trend("Ad") + season("M"))
```

```
# fit, inspect, compare
fit <- train |> model(snaive = SNAIVE(y),
  hw_mult = ETS(y ~ error("M") +
    trend("A") + season("M")))
fit |> select(hw_mult) |> report()
fit |> select(hw_mult) |> components() |>
  autoplot()
fc <- fit |> forecast(h = 12)
fc |> autoplot(df, level = 80)
fc |> accuracy(df) |> arrange(MASE)
fit |> select(hw_mult) |> gg_tsresiduals()
```

Four Ways These Methods Go Wrong

More components mean more ways to over-fit and more ways to extrapolate badly.

A linear trend over a long horizon	The straight line keeps growing at today's rate for as long as you ask it to. Past a year or two, that is rarely credible. Damp it.
The wrong seasonal form	Additive seasonality on a growing multiplicative series under-forecasts every future peak — as the accuracy table showed.
Too little history for the season	Holt-Winters needs at least two full seasonal cycles to initialise, and more to estimate γ sensibly. With seven months of daily data, the weekly season is fine; the annual one is not.
Adding components the data did not ask for	A trend term on a trendless series adds noise to every forecast. Fit only the candidates your diagnostics point to.